

Structural LQI–PIDF equivalence and constrained weight tuning for a wide-range Buck converter

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Abstract

Metaheuristic tuning of PID-type controllers for power converters usually searches gain spaces that carry no stability guarantee, and its results are rarely checked against switching effects. This paper shows that, for a non-ideal Buck converter, the linear–quadratic–integral (LQI) law has an exact realization as a two-degree-of-freedom PIDF that measures only the output voltage. When its derivative pole is placed on the capacitor ESR zero, the filter state reconstructs the capacitor voltage along every trajectory. The PIDF and the full-state LQI therefore coincide under saturation, anti-windup and discontinuous conduction. They differ only through an explicit output-current term, output-network mismatch and sampling, and a first-order-hold discretization reduces the sampling gap by a factor of about 40. The controller is tuned by searching the LQI weights, not the gains, so every candidate is LQ-optimal and carries guaranteed margins and a certified pole region. A constrained, multi-fidelity search includes short switched-model simulations to reject coexisting switching limit cycles that averaged models miss. On a 48 V converter commanded from 1 to 46 V, six metaheuristics converge to the same optimum. An unconstrained gain search reaches the same controller, but 42 % of its feasible solutions are not LQ-optimal. On a switched replica of the reference Simulink model, the tuned controller eliminates overshoot and respects the device current rating, whereas standard PI, PID, PIDF, LQR, LQG and LQI designs exceed the specification by factors of 6 to 30.

Keywords: Buck converter, LQI control, 2-DOF PIDF, structural equivalence, metaheuristic optimization, multi-fidelity constraints, switching limit cycles

1. Introduction

Most DC–DC converters in industrial equipment are still closed by a PID-type loop. The reason is practical: a PID block is available in every digital controller, technicians know how to read its three gains, and it needs a single voltage sensor. Model-based laws such as the linear–quadratic regulator are better understood mathematically, but they ask for more sensors and their tuning knobs, the weights of a quadratic cost, have no obvious meaning on the shop floor. ISA Transactions addresses the engineering of measurement and automation for practitioners and applied researchers, and this gap between what theory certifies and what industry implements is the subject of this paper. We study one converter in detail, a non-ideal asynchronous Buck stage fed from 48 V and commanded from 1 V to 46 V, and ask a precise question: can a controller be tuned in the coordinates of optimal control and still be delivered to the plant as an ordinary PID with a derivative filter, without losing the optimality certificate on the way?

The answer turns out to be yes, and exactly so, under conditions that we state and test. The rest of this section reviews how PID-type and linear–quadratic controllers reached their present form and why the non-ideal Buck is a demanding test case. It then states three research gaps, each paired with the design choice it motivates (the PID equivalent, the CMO–LQI formulation, the metaheuristics and refineBO), followed by the working assumptions, the difficulties and the contributions.

1.1. Literature review

1.1.1. From the three-term controller to the 2-DOF PIDF

The three-term law was first analysed by Minorsky [45] for the automatic steering of ships, and the rules published by Ziegler and Nichols [70] made it an industrial standard. On a Buck converter the reason for each term is easy to see. With proportional action alone the output settles below its reference whenever the load draws current, because only a standing error can pull the duty ratio away from its feedforward value. Integral action removes that offset, which gives the PI controller. The output filter, however, is lightly damped (in our model $\zeta \approx 0.14$ at 24 V), and a PI has no phase lead to offer it; the derivative term supplies that lead and turns the PI into a PID [6]. An ideal derivative cannot be built, and on a converter it would also amplify the switching ripple riding on the measured voltage. Industrial controllers and

the Simulink PID block therefore filter it with a first-order pole N , and the PID becomes a PIDF [4, 6].

The PIDF still uses one error signal for two different jobs. The gains that make it recover quickly from a load step act in the same way on every change of reference, so a loop tuned for load rejection tends to overshoot on a large setpoint step, such as the 1→46 V step of our mission. Setpoint weighting decouples the two. The proportional and derivative terms act on $br - y$ and $cr - y$ instead of $r - y$, while the integral keeps acting on the true error, so the steady state does not change [6, 5]; Bianchi et al. [9] extended the idea to multivariable PID loops. The general principle goes back to Horowitz [27]: one path of the controller is designed for robustness and disturbance rejection, and a second path, driven by the reference alone, shapes the command response. Setpoint weighting is the simplest instance of this two-degree-of-freedom (2-DOF) structure. Adding N , b and c to the three gains gives a law with six parameters, (K_p, K_i, K_d, N, b, c) , which is as far as a standard PID block goes; we take it as the end point of the upper branch of Fig. 1. Surveys of PID practice document both the dominance of this family in industrial loops and the very large number of tuning methods proposed for it, a sign that tuning, more than structure, is where the difficulty lies [5, 4].

1.1.2. From LQR to LQI

The lower branch of Fig. 1 starts with Kalman [32], who posed the regulation of a linear system as the minimization of a quadratic cost and showed that its solution is a constant state feedback computed from a Riccati equation. The LQR comes with a certificate that no PID tuning rule offers: for a single input, the return difference satisfies $|1 + L(j\omega)| \geq 1$, which gives an infinite upward gain margin, a gain reduction margin of one half and at least 60° of phase margin [33, 57, 3].

The certificate assumes that every state is measured. When the states are estimated by a Kalman filter, the controller becomes the LQG law [8], and Doyle [19] showed with a counter-example that LQG loops can have arbitrarily small margins; loop-transfer recovery restores them only at the cost of observer bandwidth [20]. The second limitation of the LQR is that a proportional state feedback does not remove the steady error caused by a constant load or by a model error. The servomechanism theory of Young and Willems [69] and Davison [15], grounded on the internal model principle of Francis and Wonham [23], solves this by adding the integral of the tracking

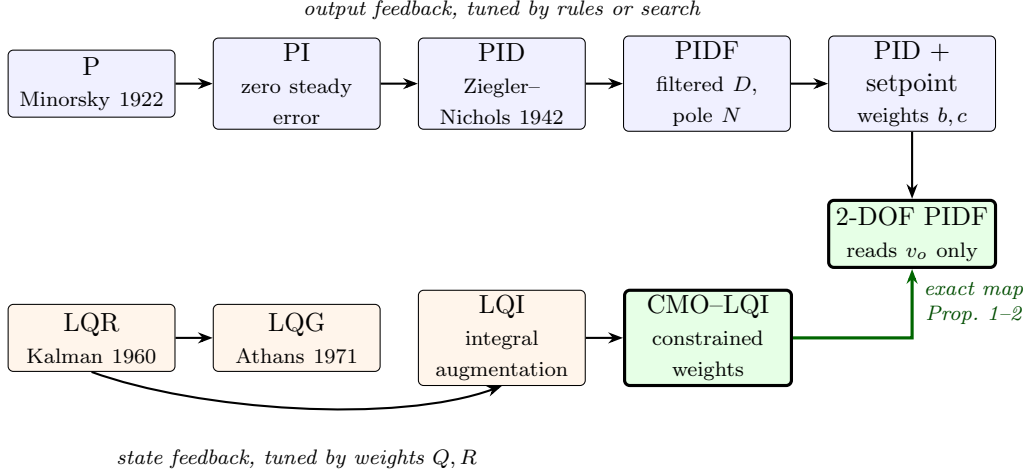


Figure 1: Evolution of the two controller families used in this paper. Upper branch: output-feedback PID-type laws, from the proportional controller to the 2-DOF PIDF. Lower branch: linear-quadratic state feedback, from LQR to LQI and to the constrained weight tuning (CMO-LQI) proposed here. The two branches meet through the exact realization of Propositions 1 and 2.

error to the state vector and solving the LQR for the augmented plant. This linear-quadratic-integral (LQI) law is what the rest of the paper uses.

Several variants of the LQI have been applied to converters. The weights have been chosen by Bryson’s rule [11], by hand, or by an optimizer; robust versions replace the Riccati equation by linear matrix inequalities that hold over a polytope of operating points [51], and closed-loop poles can be confined to a prescribed region by the same means [13]; early work already cast converter regulation as a general LQR problem [40]. Saturation of the duty ratio requires an anti-windup mechanism, for which Kothare et al. [38] gave a unified framework and back-calculation remains the usual industrial choice [7]. Other families exist and are strong on converters, for instance sliding-mode control [66], hybrid model predictive control [25, 41], and observer-based or intelligent nonlinear laws [1, 61, 10], but they give up either the fixed switching frequency or the low-order linear structure that a PID block can host. The LQI is chosen here for three reasons that are specific to the Buck: the averaged plant is second order, so the augmented state has dimension three and the Riccati solution is cheap enough to be computed inside an optimizer; the integral state gives zero steady error across a range in which the equilibrium

91 duty moves from 0.03 to 0.97; and, as shown in Section 3, the capacitor ESR
 92 places a zero in the output equation that makes the LQI exactly realizable
 93 from the output voltage.

94 1.1.3. *Why a non-ideal Buck converter*

95 The Buck converter is often used as a benchmark because its ideal aver-
 96 aged model is linear in the states and bilinear in the duty ratio [44, 21]. For
 97 the question asked here the ideal model is not enough. The ESR r_C creates
 98 the output-network zero on which the exact equivalence rests, so it cannot be
 99 dropped. The other parasitic elements matter for the operating point. Each
 100 volt lost in r_L , R_{on} , r_d or across the diode threshold V_f has to be made up
 101 by a larger duty ratio, and at a 1 V output these drops are no longer small
 102 compared with the output itself: our model gives an efficiency of about 61 %
 103 there, so nearly 40 % of the input power ends up in these elements [36, 60].
 104 The freewheeling diode adds a second effect. It blocks reverse current, so at
 105 light load or during a deep step-down the inductor current falls to zero in-
 106 side the switching period and the converter enters discontinuous conduction,
 107 where the small-signal model loses one order [65]. Our 1 V operating point
 108 sits only 3 % above the conduction boundary at the nominal load and spends
 109 15.00 % of its steady-state time in discontinuous conduction. Averaging itself
 110 is an approximation whose validity depends on the ripple and the switching
 111 frequency [39, 42], and in digital loops the sampled ripple and the duty res-
 112 olution can sustain limit cycles that no averaged model predicts [53, 52, 12].
 113 A wide-range, non-ideal Buck therefore combines saturation, a mode change
 114 and switching nonlinearities in a plant that is still simple enough to analyse
 115 in closed form, which makes it a fair test of any claim of equivalence.

116 1.2. *Research gap and motivation*

117 The review leaves three gaps open. Each one motivates a design choice
 118 of this paper, and we state them in pairs.

119 1.2.1. *Gap 1: the LQI–PID link is known only in nominal form*

120 The link between the LQI of a Buck converter and a filtered PID is known
 121 as a transfer-function identity at one operating point. Whether it survives
 122 duty saturation, anti-windup, discontinuous conduction and parameter mis-
 123 match has not been shown, yet these are exactly the conditions a converter
 124 meets across a wide range. Without that result, a PID tuned from an LQI
 125 is only an approximation of it, and the LQ margins do not carry over.

Table 1: Controllers used in this study.

Controller	Measured signals	Tuning parameters	Certificate	Role here	Key references
PI	v_o	K_p, K_i	none	comparator	Ziegler and Nichols [70]
PID	v_o	K_p, K_i, K_d	none	comparator	Åström and Hägglund [6]
PIDF	v_o	K_p, K_i, K_d, N	none	comparator	Ang et al. [4]
2-DOF PIDF	v_o	K_p, K_i, K_d, N, b, c	none if tuned directly	realization of the proposal	Horowitz [27]
LQR	i_L, v_o	Q, R	LQ margins	comparator	Kalman [32]
LQG	v_o (+ observer)	Q, R , noise covariances	none	comparator	Athans [8]
LQI	i_L, v_o	Q, R	LQ margins, zero steady error	comparator	Young and Willems [69]
CMO-LQI-PIDF	v_o	4 log-excursions	LQ margins, D-region	proposal	this work

1.2.2. Motivation 1: why a PID equivalent, and why the 2-DOF PIDF

If the LQI is better founded, one may ask why it should be turned into a PID at all. There are three reasons. The LQI needs the inductor current, hence a current sensor with its own bandwidth, cost and noise, whereas the output voltage is measured on every converter. Commissioning engineers can inspect, bound and retune three gains, and the PID block comes with validated anti-windup and bumpless transfer [6]. And tuning a PID directly in gain space, by rules or by an optimizer, gives no stability or robustness guarantee for the point it returns [4]. An exact map from the LQI to a PID-type law removes that last objection: every candidate of a search becomes a controller that has the LQR margins and still runs on a standard block.

The map needs the extra parameters of the 2-DOF PIDF. The LQI feeds back the capacitor voltage and the inductor current; replacing them by the measured output requires a filtered derivative whose pole sits on the ESR zero, and that pole is the N of the PIDF. The LQI also sees the reference only through its integral state, so the equivalent PIDF must have $b = 1$ and $c = 0$. Without setpoint weights a PIDF can match the disturbance response of the LQI but not its command response (Proposition 1). A PI or a PID lacks the parameters for either condition, which is why they appear here only as comparators. Table 1 lists all the controllers of the study.

1.2.3. Gap 2: metaheuristics tune gains, not specifications

Metaheuristic tuning of converter controllers searches gain spaces [30, 48, 59, 67]. A good result therefore carries no guarantee, the size of the search box is a guess, and the objective is usually a tracking index rather than the list of limits a specification actually contains. Searching the LQI weights instead has not been compared with a direct gain search under the same budget.

153 *1.2.4. Motivation 2: CMO-LQI, constrained multi-criteria optimization of*
 154 *the LQI weights*

155 Bryson’s rule gives a sensible starting point, but its weights are not op-
 156 timal with respect to overshoot, device current or loss. We call CMO-LQI
 157 (constrained multi-criteria optimization of the LQI) the problem of choosing
 158 the weights so that the resulting LQI, realized as a 2-DOF PIDF, minimizes
 159 an aggregate of seven performance criteria over six scenarios under hard engi-
 160 neering constraints. *Multi-criteria* is meant literally, not in the Pareto sense:
 161 the criteria are normalized and averaged into one scalar, and the constraints
 162 are handled by feasibility rules [16]. A weighted scalar index does not de-
 163 scribe a trade-off front [56], and we do not claim one. The search variables
 164 are the logarithms of the Bryson excursions and of the integral bandwidth,
 165 four numbers with a physical scale, rather than the PIDF gains.

166 *1.2.5. Gap 3: candidates are accepted on the model that selected them*

167 Most tuning studies accept a candidate on the averaged model that drives
 168 the search. In an earlier version of the present protocol, four of six winners
 169 accepted in this way failed on the switched model, because of switching limit
 170 cycles that the averaged model cannot represent. The same studies often
 171 introduce a new operator without a matched comparison against existing
 172 ones, so it is hard to tell whether a result comes from the algorithm or from
 173 the problem.

174 *1.2.6. Motivation 3: why metaheuristics, and why refineBO*

175 The CMO-LQI problem has only four variables, but its objective and
 176 its constraints come from simulations with saturation, mode changes and
 177 switching, so no gradient is available and the feasible set may be discon-
 178 nected. Population-based metaheuristics are the usual tool for such black-
 179 box problems. Two results from the optimization literature shape the way
 180 we use them. The no-free-lunch theorems state that no algorithm domi-
 181 nates on all problems [68], and metaphor-based variants are often published
 182 without evidence that they differ from existing methods [63]. We therefore
 183 do not propose a new metaheuristic. We take three classical operators with
 184 well-documented behaviour, particle swarm optimization [37, 54], the genetic
 185 algorithm [26] and the artificial bee colony [34, 35], and three recent variants
 186 built on the grey wolf optimizer and on an adaptive bee colony [46, 47, 28, 2],
 187 and ask whether the answer depends on the operator. Constraints are han-
 188 dled by Deb’s feasibility ordering, which avoids penalty factors [16, 43]; the

Table 2: Positioning with respect to representative Buck control and tuning studies. “Not reported” refers to the stated scope of each work only.

Study	Plant evidence	Controller / search	Relative to the present question
Siddhartha et al. [60]	non-ideal Buck, dSPACE	IMC-PID, direct	no repeated solver comparison, no switched energy audit
Izci et al. [30]	averaged Buck	PID, AEO + Nelder-Mead	gain-space search, no structural certificate, no switched admissibility
Nanyan et al. [48]	averaged Buck	PID, improved SCA	gain-space search, single fidelity
Siano and Citro [59]	Buck prototype	fuzzy, MO-PSO	no equal-budget solver comparison
Abdelmalek et al. [1]	Buck, experimental	observer-based nonlinear, PSO	tracking/robustness focus, not a matched co-design
Olalla et al. [51]	PWM converters, polytopic model	robust LQR	polytopic certificate, no PID realization, no search over weights
Coutinho et al. [14]	multiloop process	PID, Bayesian optimization	not a power converter
Present work	audited non-ideal Buck, 1–46 V, switched replica with energy balance	LQI weights $\rightarrow$ exact 2-DOF PIDF; 6 global operators + refineBO; 6 families	structural equivalence theorem; 33 inequalities (3 measured on the switched model); switched validation vs. PI, PID, PIDF, LQR, LQG, LQI

runs are compared on matched seeds with non-parametric tests [18, 17]; and three of the 33 constraints are measured on a short switched simulation inside every evaluation, which answers the acceptance problem of Gap 3.

A global operator spends most of its budget exploring. Once a promising region is found, a local model-based stage is more economical, which is the idea of memetic and hybrid schemes [49]. For expensive black-box functions, Bayesian optimization fits a Gaussian-process surrogate to the evaluated points and chooses the next one by maximizing an acquisition function such as the expected improvement [31, 55, 62, 58]. Unknown constraints are handled by multiplying the acquisition by the modelled probability of feasibility [24], and a trust region keeps the surrogate local and reliable [22]. Bayesian optimization has already been used to tune industrial controllers [50] and digital PID loops [14]. The refineBO stage used here combines these three elements: it starts from the history of each global run, models the objective and all 33 constraints with Gaussian processes, and searches a shrinking region around the best feasible records.

Table 2 positions the work against representative Buck control and tuning studies. The motivation, in short, is to close the three gaps on a plant that is fully specified, so that every number in the paper can be recomputed.

1.3. Assumptions

The analysis rests on the following assumptions, each of which is tested or bounded later in the paper.

- 211 • The input voltage is constant, $V_{in} = 48\text{ V}$; line regulation is not part of
212 the specification.
- 213 • The output network (R, r_C, C) is known at design time. Load steps are
214 treated as unmeasured disturbances, and the error they cause in the
215 equivalence is given in closed form.
- 216 • The PIDF measures the output voltage only. The inductor current is
217 measured only by the full-state comparators and by the state-feedback
218 twin of the proposal.
- 219 • Control is sampled synchronously with the 50 kHz carrier, with a one-
220 sample computational delay; ADC quantization and the finite resolu-
221 tion of the PWM are not modelled.
- 222 • Losses are the conduction and snubber losses of the declared model
223 boundary; switching, reverse-recovery, magnetic and thermal losses are
224 outside it.
- 225 • Component tolerances were not supplied. Robustness claims are made
226 on the nominal operating polytope only.

227 1.4. Challenges

228 Three difficulties had to be solved. The first is theoretical: showing that
229 the equivalence holds along trajectories, not only in the frequency domain,
230 and identifying exactly which disturbances break it. The second is numeri-
231 cal. A zero-order-hold discretization of the derivative filter, the usual choice,
232 introduces a lag that separates the sampled PIDF from the sampled LQI by
233 about 14.54 V during a wide slew, so the discretization had to be derived from
234 the observer interpretation of the filter. The third concerns the search. The
235 feasible set is defined by 33 inequalities, three of which depend on switching
236 phenomena; checking them on the switched model is expensive, while ignor-
237 ing them lets through candidates that are unacceptable on the real power
238 stage. The screen had to be short enough to run inside every evaluation and
239 long enough to excite the coexisting limit cycles.

240 1.5. Contributions

241 The contributions of the paper are the following.

- 242 1. A formula-first chain from the non-ideal converter equations to the
243 exact output-feedback realization of any LQI law as a 2-DOF PIDF,
244 with every analytical quantity (equilibria, transfer functions, Riccati
245 gain, LQR margins, a common D-region certificate over the operating
246 polytope) instantiated and independently recomputed.
- 247 2. A structural equivalence theorem (Proposition 2): the PIDF filter state
248 equals the capacitor voltage along every trajectory. Its domain of va-
249 lidity is stated, the gap to the full-state law is given in closed form, and
250 a first-order-hold discretization reduces the sampled gap from 14.54 V
251 to 0.35 V.
- 252 3. A constrained multi-criteria formulation (CMO-LQI) in the space of
253 LQI weights, with 33 inequalities in engineering units, three of which
254 are measured on the switched model during the search, together with
255 the evidence from two pilots of why an analytical ripple bound or a
256 steady switched screen is not sufficient.
- 257 4. A matched, equal-budget comparison of seven algorithms (six global
258 operators followed by refineBO) across six controller families. It shows
259 that the LQI coordinates do not speed up the search but confine it, at
260 no cost in optimality, to the LQ-optimal set on which the gain-space
261 optimum also lies.
- 262 5. A switched-model validation on the 24→1→46 V mission against PI,
263 PID, PIDF, LQR, LQG and LQI, reported specification by specifica-
264 tion.

265 1.6. Paper organization

266 Section 2 derives the non-ideal model and describes the switched replica
267 of the reference Simulink model. Section 3 states the LQI synthesis and
268 proves the nominal and structural equivalences. Section 4 defines the CMO-
269 LQI problem and the seven algorithms. Section 5 reports the verification
270 of the equivalence, the algorithm and family comparisons, and the switched
271 validation, and discusses the limitations. Section 6 concludes. The code,
272 including a MATLAB S-function that runs the same control layer inside the
273 reference Simulink model, is released with the paper.

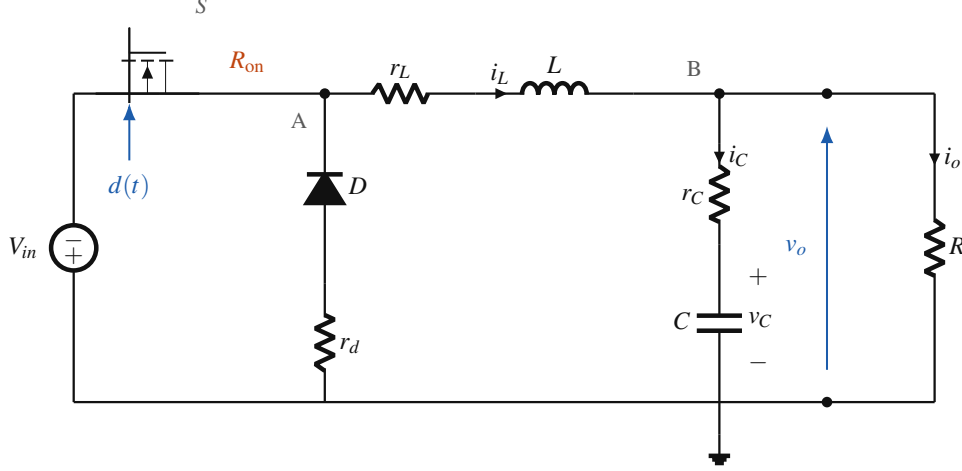


Figure 2: Non-ideal asynchronous Buck converter and adopted conventions.

274 2. Non-ideal Buck converter: from the circuit to the switched 275 replica

276 2.1. Circuit and conventions

277 Fig. 2 shows the asynchronous converter. The state is $x = [i_L \ v_C]^\top$; i_L
278 flows from the switching node to the output, i_C flows downward in the capac-
279 itor branch and the load disturbance i_p is positive when it draws additional
280 current. The source is fixed, $V_{in} = 48 \text{ V}$. The high-side switch is an IRF540N
281 represented by its static on-resistance R_{on} ; the freewheel diode follows the
282 affine conduction law of the STPST10H100SF data sheet, $v_D = V_f + r_d i_D$
283 [29, 64]. Table 3 lists the values, all decoded from the workspace of the
284 reference Simulink model.

285 2.2. Averaged model and loss-aware equilibrium

286 The output node and the ESR give

$$i_L = i_C + \frac{v_o}{R} + i_p, \quad v_o = v_C + r_C i_C, \quad (1)$$

287 hence

$$i_C = \frac{R(i_L - i_p) - v_C}{R + r_C}, \quad v_o = \frac{R}{R + r_C} v_C + \frac{R r_C}{R + r_C} (i_L - i_p). \quad (2)$$

Table 3: Electrical and numerical parameters (reference model workspace).

Quantity	Value	Role
V_{in}	48 V, constant	source
L, r_L	100 μ H, 0.10 Ω	inductor branch
C, r_C	100 μ F, 0.05 Ω	capacitor branch
R	10 Ω	nominal load
R_{on}	0.05 Ω	IRF540N static surrogate
V_f, r_d	0.42 V, 0.020 Ω	STPST10H100SF affine law
R_s ($C_s = \infty$)	100 k Ω	resistive snubber across the switch
$R_{d,body}$	0.01 Ω	MOSFET body diode
f_s	50 kHz	PWM carrier
T_s	10 μ s	controller sampling ($= 1/(2f_s)$)
$[d_{\min}, d_{\max}]$	[0.02, 0.98]	total duty saturation

288 With the switch on, $L\dot{i}_L = V_{in} - (r_L + R_{on})i_L - v_o$; with the diode conducting,
 289 $L\dot{i}_L = -V_f - (r_L + r_d)i_L - v_o$; in both cases $C\dot{v}_C = i_C$. Duty averaging yields

$$L\dot{i}_L = dV_{in} - [r_L + dR_{on} + (1-d)r_d]i_L - (1-d)V_f - v_o. \quad (3)$$

290 At equilibrium $I_L = V_o/R + I_p$ and the loss-aware duty is

$$d_{ff} = \frac{V_o + (r_L + r_d)I_L + V_f}{V_{in} - (R_{on} - r_d)I_L + V_f}. \quad (4)$$

291 Every controller of this paper uses (4) with nominal parameters and $I_p = 0$
 292 as feedforward; the feedforward never sees the simulated plant.

293 2.3. Small-signal model and transfer functions

294 Let $\gamma = R/(R + r_C)$, $\rho_0 = r_L + r_d + d_0(R_{on} - r_d)$ and $E_d = V_{in} +$
 295 $V_f - (R_{on} - r_d)I_{L,0}$. Linearizing (2)–(3) gives $\delta\dot{x} = A\delta x + B_d\delta d + B_p\delta i_p$,
 296 $\delta v_o = C_y\delta x + D_p\delta i_p$ with

$$A = \begin{bmatrix} -(\rho_0 + \gamma r_C)/L & -\gamma/L \\ \gamma/C & -\gamma/(RC) \end{bmatrix}, \quad B_d = \begin{bmatrix} E_d/L \\ 0 \end{bmatrix}, \quad C_y = [\gamma r_C \quad \gamma], \quad (5)$$

297 $B_p = [\gamma r_C/L \quad -\gamma/C]^\top$ and $D_p = -\gamma r_C$. The control-to-output transfer
 298 function is

$$G_{vd}(s) = \frac{E_d R(1 + r_C C s)}{\Delta(s)}, \quad (6)$$

$$\Delta(s) = LC(R + r_C)s^2 + [L + C((R + r_C)\rho_0 + R r_C)]s + R + \rho_0,$$

Table 4: Equilibria, monic $\Delta(s)$, G_{vd} numerator and stationary conduction screen at the three commanded levels (nominal load, $i_p = 0$). $\omega_z = 1/(r_C C) = 2 \times 10^5$ rad/s at every level.

V_o (V)	$I_{L,0}$ (A)	d_0	ρ_0 (Ω)	E_d (V)	$\Delta(s)/(LC(R + r_C))$	G_{vd} numerator	$\Delta I_{L,pp}$ (A)	I_{crit} (A)	$I_L - I_{crit}$ (A)	R_{crit}/R
1	0.1	0.029576	0.120887	48.417	$s^2 + 2701.41s + 1.00705 \times 10^8$	$2.40881 \times 10^4 s + 4.81761 \times 10^9$	0.1941	0.0970	0.0030	1.030
24	2.4	0.511045	0.135331	48.348	$s^2 + 2845.85s + 1.00849 \times 10^8$	$2.40537 \times 10^4 s + 4.81075 \times 10^9$	2.3470	1.1735	1.2265	2.045
46	4.6	0.972868	0.149186	48.282	$s^2 + 2984.40s + 1.00987 \times 10^8$	$2.40209 \times 10^4 s + 4.80418 \times 10^9$	0.2496	0.1248	4.4752	36.856

with the ESR zero $\omega_z = N_C = 1/(r_C C) = 2 \times 10^5$ rad/s. Table 4 instantiates (4)–(6) at the three commanded levels; the closed form and a state-space conversion agree to machine precision. At the 24 V synthesis point, (A, B_d) and (A, C_y) have rank two and the integral-augmented pair has rank three.

2.4. Conduction mode across the range

For a duty d_{ff} the inductor ripple, the boundary current and the boundary load are

$$\Delta I_{L,pp} = \frac{V_o(1 - d_{ff})}{Lf_s}, \quad I_{crit} = \frac{\Delta I_{L,pp}}{2}, \quad R_{crit} = \frac{2Lf_s}{1 - d_{ff}}. \quad (7)$$

The stationary screen (Table 4) is positive at every level but not uniform: at 1 V the current margin is 2.96 mA (3.05 % of I_{crit}) and R_{crit} is only 1.030 times the nominal load. The switched model is more pessimistic than this screen: over the steady 1 V window it spends 15.00 % of the time in discontinuous conduction at 60.57 % efficiency, because the diode drop is 42 % of the commanded output. Discontinuous conduction is thus a nominal regime of the mission, and every controller is validated in it.

2.5. Switched replica of the reference model

The acceptance model (Fig. 3) is the Simulink/Simscape Electrical model `buck_fo_lqi_fixed_reference_v6.1.46V_R2022a.slx`. For the campaign we use a replica written from its block parameters (`cmo/sim.py`): a resistive switch with snubber R_s and a body diode $R_{d,body}$; the affine freewheel diode, which blocks reverse current so that discontinuous conduction arises by itself (the 1 ns snubber-leakage branch is solved quasi-statically); a continuous sawtooth carrier whose edges are located exactly inside the integration step; a controller sampled at T_s with a one-sample output delay, exactly as the S-function of the reference model. The step is 0.2 μ s (100 steps per switching period) with fourth-order Runge–Kutta integration.

324 The energy boundary is the one declared in the model workspace:

$$\begin{aligned} P_{in} &= V_{in}i_{in}, & P_u &= \frac{v_o^2}{R} + v_o i_p, \\ P_{loss} &= r_L i_L^2 + r_C i_C^2 + v_{SW} i_{SW} + v_D i_D, \end{aligned} \quad (8)$$

325 the snubber dissipation being contained in $v_{SW}i_{SW}$. The three energies are
 326 integrated with the same Runge–Kutta stages as the states, so the residual
 327 $E_{in} - E_u - E_{loss} - \Delta E_{stored}$ measures integration consistency only. It stays
 328 below $4.5 \times 10^{-6} \%$ of the input energy for every replay reported here.

329 The replica reproduces the controller-independent figures of the reference
 330 model: 15.00 % of discontinuous conduction and 60.57 % efficiency in the 1 V
 331 window (15.27 % and 60.51 % in the Simulink audit of the previous version
 332 of this study). Its discrete LQI comparator gain, synthesized by the rule
 333 of Section 4.5, equals the gain stored in the model workspace to all printed
 334 digits, and the fixed PI and LQI comparators reproduce the Simulink over-
 335 shoot and peak current to within 2 % (Section 5.5). A MATLAB S-function
 336 (`matlab/cmo_sfun.m`) implements the same control layer line by line, and
 337 `matlab/run_cmo_validation.m` replays every frozen controller on the refer-
 338 ence `.slx` without modifying it.

339 3. LQI synthesis in weight space and its exact 2-DOF PIDF real- 340 ization

341 3.1. Integral augmentation and Bryson weights

342 Let $z(t) = \int_0^t (r - v_o) d\tau$. Around the synthesis equilibrium ($V_o = 24$ V)
 343 the augmented plant is

$$\dot{x}_a = A_a x_a + B_a \delta d, \quad A_a = \begin{bmatrix} A & 0 \\ -C_y & 0 \end{bmatrix}, \quad B_a = \begin{bmatrix} B_d \\ 0 \end{bmatrix}, \quad x_a = \begin{bmatrix} \delta x \\ z \end{bmatrix}, \quad (9)$$

344 and the LQI law $\delta d = -K x_a$, $K = [K_{ix} \ K_{vx} \ K_z]$, minimizes $\int_0^\infty (x_a^\top Q x_a +$
 345 $R_d \delta d^2) dt$ with

$$A_a^\top P + P A_a - P B_a R_d^{-1} B_a^\top P + Q = 0, \quad K = R_d^{-1} B_a^\top P. \quad (10)$$

346 Bryson’s rule assigns $Q = \text{diag}(\Delta_i^{-2}, \Delta_v^{-2}, (\Delta_v/\omega_B)^{-2})$ and $R_d = \Delta_d^{-2}$: Δ_i
 347 and Δ_v are admissible current and voltage excursions and the integral state,
 348 in V s, is scaled by a target bandwidth ω_B . The duty excursion is fixed at

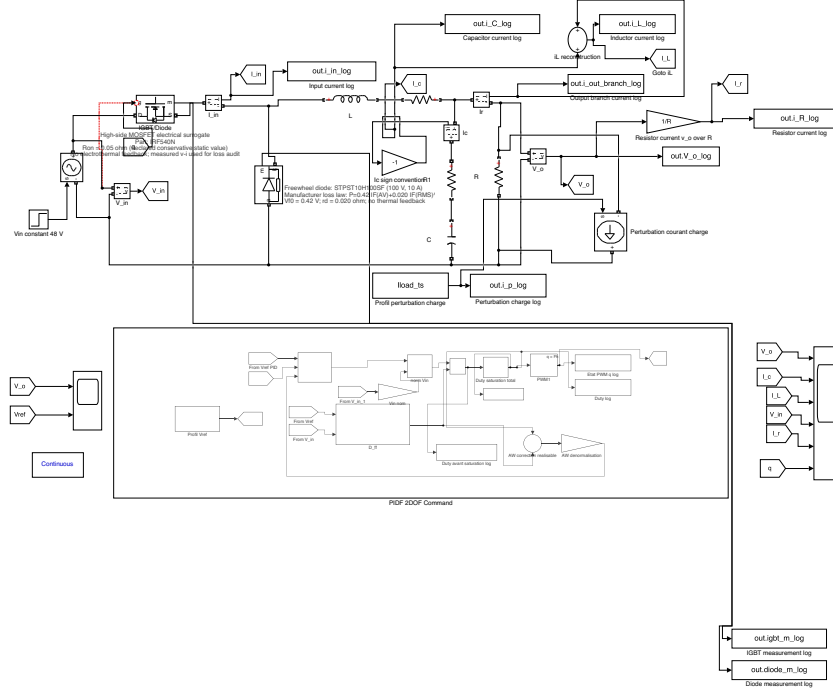


Figure 3: Reference Simulink/Simscape Electrical switching model (fixed 48 V source, IRF540N surrogate, STPST10H100SF diode, resistive snubber, continuous PWM at 50 kHz, sampled controller subsystem and energy loggers). Only the controller S-function is replaced when a candidate is replayed.

its physical value $\Delta_d = (d_{\max} - d_{\min})/2 = 0.48$, which removes the scale redundancy $(Q, R_d) \rightarrow (\alpha Q, \alpha R_d)$ of (10). The decision vector of the co-design is

$$\xi = [\log_{10} \Delta_i, \log_{10} \Delta_v, \log_{10} \omega_B, \log_{10} K_b]^\top, \quad (11)$$

where K_b is the back-calculation rate of the anti-windup, which lies outside the linear theory and is searched alongside the weights. The normalization record ξ_0 uses $\Delta_i = 15$ A, $\Delta_v = 46$ V (the largest commanded output) and $\omega_B = K_b = 2\pi \cdot 1500$ rad/s.

3.2. Frequency-domain realization

For $i_p = 0$ the output network gives, in the Laplace domain, $v_C = v_o/(1 + r_C C s)$ and $i_L = v_o/R + C s v_o/(1 + r_C C s)$. Substituting into the state part of

the LQI law,

$$\frac{K_{ix}i_L + K_{vx}v_C}{v_o} = \frac{K_{ix}}{R} + \frac{K_{ix}Cs + K_{vx}}{1 + r_CCs} = K_p + K_d \frac{N_Cs}{s + N_C}, \quad (12)$$

which exposes one constant and one filtered derivative term.

Proposition 1 (Nominal realization). *The output-error compensator generated by the full-state LQI law is the 2-DOF PIDF*

$$d = d_{ff}(r) + K_p(br - v_o) + \frac{K_i}{s}(r - v_o) + K_d \frac{N_Cs}{s + N_C}(cr - v_o), \quad (13)$$

with

$$\begin{aligned} K_p &= K_{vx} + \frac{K_{ix}}{R}, & K_i &= -K_z, & K_d &= C(K_{ix} - r_CK_{vx}), \\ N &= N_C = \frac{1}{r_C C}, & b &= 1, & c &= 0. \end{aligned} \quad (14)$$

Proof. Split the right-hand side of (12) into its zero-frequency value $K_{vx} + K_{ix}/R$ and the remainder $C(K_{ix} - r_CK_{vx})N_Cs/(s + N_C)$. The equilibrium map $x_{eq}(r) = [r/R \ r]^\top$ contributes $+(K_{ix}/R + K_{vx})r = K_p r$, hence $b = 1$; the state-feedback law has no derivative of the reference, hence $c = 0$; the integral state contributes $-K_z z$, hence $K_i = -K_z$. $\square$

Because $N_C = 2 \times 10^5 \text{ rad/s} < \pi/T_s = 3.14 \times 10^5 \text{ rad/s}$, the exact filter pole is implementable at the adopted sampling period, and no projected surrogate is needed: every candidate of the search is realized with its own exact pole.

3.3. Structural (trajectory-level) equivalence

Proposition 1 is a statement about transfer functions, i.e. about the linear nominal plant with consistent initialization. The next result shows that the realization is in fact exact along every trajectory of the *nonlinear* converter, because it only uses the output network, never the inductor branch. Write the PIDF of (13) in state form,

$$\begin{aligned} \tilde{d} &= d_{ff}(r) + K_p(br - v_o) + x_I + K_d N(cr - v_o - x_F), & d &= \text{sat}(\tilde{d}), \\ \dot{x}_I &= K_i(r - v_o) + K_b(d - \tilde{d}), & \dot{x}_F &= N(cr - v_o - x_F), \end{aligned} \quad (15)$$

and the LQI with the same feedforward and anti-windup as

$$\begin{aligned} \tilde{d} &= d_{ff}(r) - K_{ix}\left(i_L - \frac{r}{R}\right) - K_{vx}(v_C - r) - K_z z, \\ \dot{z} &= (r - v_o) + \frac{K_b}{|K_z|}(d - \tilde{d}). \end{aligned} \quad (16)$$

380 **Proposition 2** (Structural equivalence). *Let r be piecewise constant, let the*
 381 *PIDF use (14) with $c = 0$, $N = N_C$ computed from the true (r_C, C) , and let*
 382 *$x_F(0) = -v_C(0)$ and $x_I(0) = -K_z z(0)$. Then, along every trajectory of the*
 383 *converter—averaged or switched, in continuous or discontinuous conduction,*
 384 *with or without active saturation and anti-windup, and for any values of L ,*
 385 *r_L , R_{on} , r_d , V_f and V_{in} :*

- 386 (i) $x_F(t) = -v_C(t)$ for all $t \geq 0$;
 387 (ii) the two unsaturated commands satisfy

$$\tilde{d}_{LQI}(t) - \tilde{d}_{PIDF}(t) = -K_{ix} \left(i_o(t) - \frac{v_o(t)}{R} \right), \quad (17)$$

- 388 where $i_o = i_L - i_C$ is the actual output-branch current and R the nominal
 389 load;
 390 (iii) if $i_o \equiv v_o/R$ (no load disturbance, nominal load), the two closed loops
 391 generate identical trajectories.

392 *Proof.* (i) The capacitor branch gives $v_o = v_C + r_C C \dot{v}_C$ exactly, whatever
 393 flows into the output node, so $\dot{v}_C = N_C(v_o - v_C)$. Let $\varepsilon = x_F + v_C$. Between
 394 reference steps, with $c = 0$, $\dot{\varepsilon} = N_C(-v_o - x_F) + N_C(v_o - v_C) = -N_C \varepsilon$, and
 395 ε is continuous across steps because $c = 0$. Hence $\varepsilon(0) = 0$ implies $\varepsilon \equiv 0$.
 396 (ii) With (i), the derivative term of (15) is $K_d N_C(v_C - v_o) = -K_d N_C r_C i_C =$
 397 $-(K_{ix} - r_C K_{vx}) i_C$. In (16) substitute $i_L = i_o + i_C$ and $v_C = v_o - r_C i_C$:
 398 $-K_{ix}(i_L - r/R) - K_{vx}(v_C - r) = K_p(r - v_o) - (K_{ix} - r_C K_{vx}) i_C - K_{ix}(i_o - v_o/R)$.
 399 With $b = 1$ and $x_I = -K_z z$ the difference of the two commands is (17). (iii) If
 400 the right-hand side of (17) vanishes, both laws issue the same $\tilde{d}$ and d at every
 401 instant. Then $\frac{d}{dt}(x_I + K_z z) = K_i(r - v_o) + K_b(d - \tilde{d}) - |K_z|(r - v_o) - K_b(d - \tilde{d}) =$
 402 0 because $K_z < 0$, so $x_I = -K_z z$ is preserved, and the two closed-loop
 403 systems coincide; uniqueness of solutions gives identical trajectories. $\square$

404 Three practical consequences follow. First, the PIDF does not approxi-
 405 mate the LQI on the nonlinear converter; it *is* the LQI whenever the load
 406 current is the nominal one, and the saturation, anti-windup and conduction-
 407 mode effects that invalidate small-signal arguments do not break it. Second,
 408 the price of removing the current sensor is stated exactly by (17): the full-
 409 state law contains an instantaneous proportional feedback $-K_{ix}(i_o - v_o/R)$
 410 on the unmodelled output current, which the PIDF must recover through its
 411 integral action. Third, the equivalence is lost by exactly three mechanisms:
 412 a load disturbance or load mismatch (through (17)), a mismatch of (r_C, C)

413 (through $N \neq N_C$ in (i)), and sampling, because the discrete filter recon-
 414 structs v_C only between samples while the sampled LQI reads i_L with its
 415 switching ripple. Section 5.2 verifies each case.

416 3.4. Guarantees and their scope

417 *LQR margins.* For the full-state augmented loop $L_K(s) = K(sI - A_a)^{-1}B_a$
 418 the return-difference identity gives $|1 + L_K(j\omega)| \geq 1$, hence the gain margin
 419 $(1/2, \infty)$ and a phase margin of at least 60° [57, 3]. By Proposition 2(iii)
 420 these margins are those of the PIDF loop at the input of the nominal linear
 421 plant; they do not survive sampling or an observer [19].

422 *Operating-polytope certificate.* The augmented closed-loop matrix is affine
 423 in $(\rho_0(d_0), E_d(I_{L,0}))$. Over $v_o \in [1, 46]$ V and $i_p \in [0, 0.5]$ A the rectangle
 424 $d_0 \in [0.0295764, 0.9744132]$, $I_{L,0} \in [0.1, 5.1]$ A has four vertices $A_{cl,k}$. A
 425 common $P \succ 0$ with

$$A_{cl,k}P + PA_{cl,k}^\top + 2\alpha P \prec 0, \quad \begin{bmatrix} \sin \theta(M_k + M_k^\top) & \cos \theta(M_k - M_k^\top) \\ \cos \theta(M_k^\top - M_k) & \sin \theta(M_k + M_k^\top) \end{bmatrix} \prec 0, \quad (18)$$

426 $M_k = A_{cl,k}P$, $\zeta = \cos \theta$, places every closed-loop matrix of the convex hull
 427 in $\{\Re s < -\alpha\} \cap \{\zeta(s) > \zeta\}$ [13]. Bisection gives the boundary values; a 1 %
 428 safety factor is applied and the recovered P is re-checked for strictness.

429 *Discrete implementation.* All controllers run at T_s with a one-sample com-
 430 putational delay, total duty saturation and forward-Euler integral and back-
 431 calculation updates. Proposition 2(i) says that the derivative-filter state is an
 432 observer of the capacitor voltage, $x_F = cr - v_C$, and it should be discretized as
 433 one. The usual zero-order-hold discretization, $x_F[k+1] = \phi x_F[k] + (1-\phi)w[k]$
 434 with $w = cr - v_o$ and $\phi = e^{-NT_s}$, uses at sample k an estimate built from
 435 $v_o[k-1]$ only; during a 45 V slew v_o moves by about 2.7 V per sample and
 436 $K_d N \approx 0.7$ turns this lag into a duty error of order one. We therefore use
 437 the triangle-hold (first-order hold) update, exact when v_o is piecewise linear
 438 between samples,

$$x_F[k] = \phi x_F[k-1] + (1-\phi)w[k-1] + \gamma(w[k] - w[k-1]), \quad \gamma = 1 - \frac{1-\phi}{NT_s}, \quad (19)$$

439 evaluated with the current sample before the command $\tilde{d}[k] = d_{ff} + K_p(br[k] -$
 440 $v_o[k]) + x_I[k] + K_d N(w[k] - x_F[k])$ is formed, and initialized with $x_F[0] =$
 441 $cr[0] - v_C(0)$. Section 5.2 shows that this single choice decides whether the
 442 sampled PIDF remains equivalent to the sampled LQI.

443 3.5. Numerical instantiation

444 At the 24 V point, (5) evaluates to

$$\begin{aligned} A &= \begin{bmatrix} -1850.8259 & -9950.2488 \\ 9950.2488 & -995.0249 \end{bmatrix} \text{s}^{-1}, \\ B_d &= \begin{bmatrix} 483480 \\ 0 \end{bmatrix}, \quad C_y = [0.0497512 \quad 0.9950249]. \end{aligned} \quad (20)$$

445 For ξ_0 , $Q = \text{diag}(0.00444444, 0.00047259, 41978.4686)$, $R_d = 4.34027778$
 446 and (10) gives $K = [0.03564341 \quad 0.01212576 \quad -98.34550916]$ with a rel-
 447 ative Riccati residual of 2.18×10^{-14} and closed-loop poles -4243.01 and
 448 $-7917.86 \pm j6986.57 \text{ rad/s}$. The realization (14) gives

$$\begin{aligned} \theta_0 &= [K_p, K_i, K_d, N, b, c, K_b] \\ &= [0.0156901028, 98.3455092, 3.5037119 \times 10^{-6}, 2 \times 10^5, 1, 0, 9424.778], \end{aligned} \quad (21)$$

449 with a zero coefficient residual and a maximum relative frequency-domain
 450 error of 8.67×10^{-16} over $[10^{-1}, 10^8] \text{ rad/s}$. The minimum return difference is
 451 1.000000 and the phase margin of the full-state loop is 87.04° . The operating-
 452 polytope certificate at ξ_0 is $\alpha_{cert} = 4157.9286 \text{ rad/s}$ and $\zeta_{cert} = 0.7330911$
 453 ($\lambda_{\min}(P) > 0$ and strictly negative LMI eigenvalues were re-verified on the
 454 recovered matrices).

455 4. Constrained co-design problem and algorithms

456 4.1. Workflow

457 Fig. 4 summarizes the protocol. Offline, a candidate ξ is mapped by (10)
 458 and (14) to a 2-DOF PIDF; the averaged evaluator runs six scenarios and
 459 the switched screen runs one 34 ms transition sequence; Deb ordering ranks
 460 the record. After 240 global and 60 local calls the best record of each run is
 461 frozen, and only then replayed on the full switched mission, which is never
 462 visible to the search.

463 4.2. Scenarios and objective

464 The averaged evaluator integrates (3) with the hybrid guard $i_L \geq 0$ (RK4,
 465 $2 \mu\text{s}$) under the sampled control layer of Section 3. Six scenarios are declared
 466 (Table 5); each starts from the analytical equilibrium preceding its first event.
 467 For scenario s the seven criteria are the loss intensity E_{loss}/E_u , IAE, ISE,

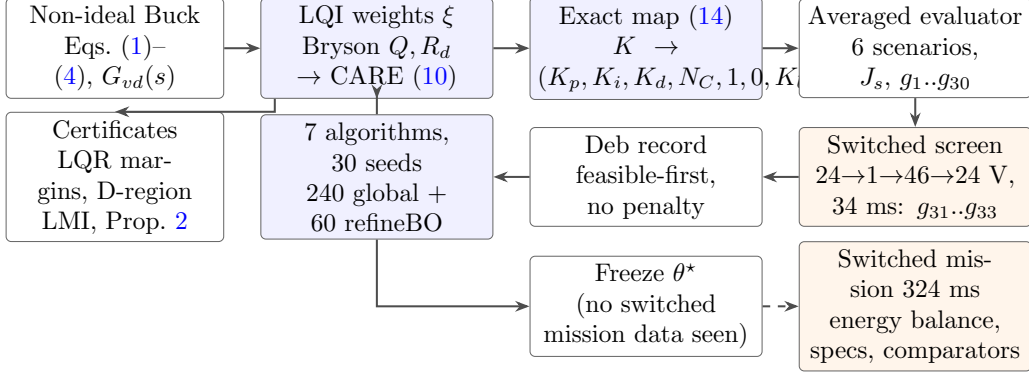


Figure 4: Formula-first CMO-LQI-PIDF workflow. Solid arrows: evaluation inside the search; dashed arrow: acceptance replay after the freeze. The switched screen is the only switched information used during the search.

Table 5: Averaged scenarios (the source is 48 V in all of them).

ID	T (ms)	reference	load pulse (+0.5 A)
S0	90	24→1→46 V at 30, 60 ms	15–25 ms
S1a	40	24→1 V at 10 ms	–
S1b	40	1→46 V at 10 ms	–
S1c	40	46→24 V at 10 ms	–
S2a	40	24 V	10–20 ms
S2b	40	46 V	10–20 ms

468 global ITAE, event-reset ITAE, RMSE and MAE, each divided by its value
 469 for the normalization record ξ_0 . The scalar objective is the mean of the 6×7
 470 ratios,

$$J_s(\xi) = \frac{1}{42} \sum_{s=1}^6 \mathbf{1}^\top r_s(\xi), \quad J_s(\xi_0) = 1. \quad (22)$$

471 MAE is IAE divided by the horizon and RMSE is the square root of ISE
 472 divided by it, so two of the seven entries are redundant; the induced weights
 473 (2/7 on the IAE family, 2/7 on the ISE family, 1/7 on each other criterion)
 474 are declared rather than left implicit. The objective ranks feasible candidates
 475 only.

476 *4.3. Thirty-three inequalities in engineering units*

477 For each scenario five inequalities $g_{s,j} \leq 0$ are evaluated separately:

$$\begin{aligned}
g_{s,1} &= \frac{\max_t \sigma_k (v_o - r_k)}{1.5 \text{ V}} - 1 && \text{overshoot beyond each new level,} \\
g_{s,2} &= \frac{\max_{t \in \mathcal{T}_p} |r - v_o|}{0.5 \text{ V}} - 1 && \text{load-pulse deviation,} \\
g_{s,3} &= \frac{\max_t i_L + \Delta I_{L,pp}^{\max}/2}{23 \text{ A}} - 1 && \text{device current,} \\
g_{s,4} &= \frac{\max_k |\bar{e}_k|}{25 \text{ mV}} - 1 && \text{steady error,} \\
g_{s,5} &= \frac{\rho_{\ell,s} - 1}{0.005} - 1, \quad \rho_{\ell,s} = \frac{E_{\text{loss}}/E_u}{(E_{\text{loss}}/E_u)_{\xi_0}} && \text{loss intensity,}
\end{aligned} \tag{23}$$

478 where σ_k is the sign of the k -th reference step, $\mathcal{T}_p$ runs from the onset of
479 the load pulse to the next event, $\bar{e}_k$ is the mean error over the last 2 ms of
480 each constant segment and 23 A is the IRF540N continuous drain rating at
481 100 °C. The averaged peak current is augmented by the worst half ripple
482 $\Delta I_{L,pp}^{\max}/2 = 1.17 \text{ A}$ (Table 4). A family that does not apply to a scenario (no
483 step, no load pulse) is set to -1 . The loss-intensity gate keeps the design
484 loss-aware: no scenario may convert energy less efficiently, per useful joule,
485 than the Bryson record ξ_0 by more than 0.5 %. These 30 inequalities are
486 completed by three measured ones,

$$g_{31} = \frac{\max_k \text{pp}(d)}{0.20} - 1, \quad g_{32} = \frac{\max_k |\bar{e}_{sw,k}|}{25 \text{ mV}} - 1, \quad g_{33} = \frac{\max_t i_{L,sw}}{23 \text{ A}} - 1, \tag{24}$$

487 obtained from one run of the switched replica over the transition sequence
488 $24 \rightarrow 1 \rightarrow 46 \rightarrow 24 \text{ V}$ (steps at 2, 14 and 24 ms, 34 ms in total, 1 μs RK4 with
489 exact PWM edges). The duty peak-to-peak and the mean error are measured
490 over the last 4 ms of each level. A candidate is feasible when $\sum_j \max(g_j, 0) \leq$
491 10^{-4} ; the screen costs about 12 ms per candidate against 14 ms for the six
492 averaged scenarios.

493 *Why the screen contains large transitions.* The protocol was frozen after two
494 pilot campaigns that are reported, not hidden. Pilot 1 used the analytical rip-
495 ple bound of the earlier version of this study, $\rho_{\text{rip}} = \max_V |C_{fb}(z=-1)| \Delta V_{o,pp}(V)/2 \leq$
496 0.20, where C_{fb} is the discrete feedback path evaluated at the controller
497 Nyquist frequency, which coincides with the carrier frequency because $T_s =$

498 $1/(2f_s)$. Every branch winner it produced satisfied the bound ($\rho_{rip} \approx 0.07$ –
 499 0.19) and yet, on the switched model, several sustained a limit cycle with
 500 duty swings of 0.7 – 0.9 at 24 V and offsets of 0.3 – 1.8 V. Pilot 2 replaced the
 501 bound by steady 10 ms switched windows started *from equilibrium* at 1 , 24
 502 and 46 V. Its best design (178 of 180 runs feasible) passed that screen, but on
 503 the reference mission it entered, after the large $24 \rightarrow 1$ V transition, a different
 504 limit cycle in which one sample in four clips at $d_{\min}$, leaving a -42 mV offset
 505 at 1 V. The loop thus holds *coexisting* switching limit cycles, and the one
 506 reached depends on the transient that precedes it; this is why the final screen
 507 is a transition sequence. ρ_{rip} is kept as a reported diagnostic only. Both pi-
 508 lots used the zero-order-hold filter; the triangle-hold update (19), adopted
 509 after the equivalence audit of Section 5.2, removes most of the limit-cycling
 510 designs by itself.

511 *Useful-energy band.* The earlier protocol also required $|E_u/E_{u,\xi_0} - 1| \leq 0.5\%$
 512 in every scenario. We report this quantity but no longer constrain it. An-
 513 chored to ξ_0 , it penalizes the removal of the 9.8 V undershoot of ξ_0 on the
 514 $46 \rightarrow 24$ V step, because a controller that holds 24 V delivers more energy than
 515 one that undershoots; anchored to the ideally regulated task, it penalizes
 516 every controller by 1 – 5% on step-down scenarios, where the asynchronous
 517 converter can only discharge through its load. With absolute overshoot, load-
 518 deviation and steady-error gates the delivered task is already pinned, and the
 519 loss intensity is normalized per useful joule, so under-delivery cannot improve
 520 it.

521 Feasibility is stated in absolute units, not as non-inferiority against ξ_0 .
 522 With ratio gates, 24 of 36 constraints are active at ξ_0 and a gentler normaliza-
 523 tion record leaves no candidate that dominates it; with absolute gates every
 524 controller family of Section 4.5 is judged against the same specification, and
 525 ξ_0 itself is infeasible because it overshoots by 6.2 V on the $1 \rightarrow 46$ V step.

526 4.4. Algorithms and budgets

527 Constraint handling uses Deb’s rules [16]: feasible beats infeasible, fea-
 528 sible records are ranked by J_s , infeasible ones by aggregate violation; no
 529 penalty weight is tuned. All methods work in the normalized box $u \in [0, 1]^n$,
 530 receive the matched seed $2026091200 + k$, $k = 1, \dots, 30$, and the same initial
 531 population of 12: ξ_0 first, six further individuals within a normalized radius
 532 0.05 of it, five uniform. One counted call is one complete evaluation (six

Table 6: Algorithms 1–7. Population 12, 240 global calls, then 60 refineBO calls initialized from the same run’s history (300 calls per run).

Alg.	Exploration / update	Exploitation / survival	Fixed parameters
1 PSO [37]	velocity with independent cognitive and social draws	personal and global Deb incumbents	$w = 0.9 - 0.5\tau$, $c_1 = c_2 = 1.7$, $ v \leq 0.2$
2 GA	binary Deb tournaments, SBX crossover	polynomial mutation, elitist ($\mu + \lambda$)	$\eta_c = 15$, $\eta_m = 20$, $p_m = 1/n$
3 ABC [35]	one-coordinate differential move (employed, onlooker)	greedy Deb replacement, scout	limit $\max(5, \text{round}(0.5 n_p n))$
4 pAEABC	ABC move scaled by diversity $s_k = \text{clip}(0.5 + 0.15/D_k, 0.5, 2)$, extra coordinate kick w.p. $0.25(1 - \tau)$	as ABC	inspired by Alsamia et al. [2]
5 pIGWO	three-leader GWO with $a = 2(1 - \tau^2)$, Gaussian kick w.p. $0.15(1 - \tau) + 0.02$	elitist pooling	inspired by Hou et al. [28]
6 pIGWO–DLH	linear $a = 2 - 2k/K$ GWO trial and an independently evaluated radial local-hunt trial	better trial replaces the wolf (non-elitist)	$R_i = \ x_i - x_\alpha\ /2$; inspired by Nadimi-Shahraki et al. [47]
7 refineBO	shared-kernel Matérn-5/2 GPs on J_s and $g_1, \dots, g_{33}$; pool of 1000 around the 10 Deb-best records, radius $0.04 \rightarrow 0.003$	$\alpha = P_f(\text{EI} + 0.02\sigma_J) + 10^{-4}P_f$, $P_f = \prod_j \Phi(-\mu_j/\sigma_j)$; restoration mode if no feasible incumbent	60 calls

533 averaged scenarios, the switched screen, 33 inequalities). Table 6 lists the
534 operators.

535 The prefix “p” marks project variants that are inspired by, but not
536 claimed to be operator-identical with, the cited algorithms. τ is the fraction
537 of the budget consumed. The product P_f of marginal feasibility probabili-
538 ties is an independence approximation used for ordering only; every accepted
539 point is checked by the exact evaluator. Wall time is logged but never inter-
540 preted as an energy or performance quantity.

541 4.5. Controller families and fixed-rule comparators

542 To separate the value of the parameterization from that of the optimizer,
543 the identical protocol (evaluator, 33 inequalities, 7 algorithms, 30 seeds, 300
544 calls) is run on six families:

- 545 • **CMO–LQI–PIDF**: ξ of (11) $\rightarrow$ (10) $\rightarrow$ (14); 4 variables, reads v_o
546 only;
- 547 • **CMO–LQI–SF**: the same ξ and the same continuous gain, applied as
548 full-state LQI with measured i_L and i_C (the twin of Proposition 2);
- 549 • **PIDF-direct**: the same 2-DOF structure ($b = 1$, $c = 0$, $N = N_C$) with
550 $(\log K_p, \log K_i, \log K_d, \log K_b)$ searched directly over the gain ranges
551 spanned by the LQI box;

- 552 • **PIDF7-direct**: $K_p, K_i, K_d, N, b, c, K_b$ all free (7 variables);
- 553 • **PID-direct**: backward-difference derivative on $-v_o$, $b = 1$, (K_p, K_i, K_d, K_b)
554 free;
- 555 • **PI-direct**: (K_p, K_i, K_b) free.

556 *Fixed-rule comparators.* Six classical designs share the control layer and are
557 synthesized at 24 V, $T_s = 10 \mu\text{s}$, with the delay $1.5T_s$ (computation plus
558 ZOH) in every frequency-domain rule. *PI*: crossover $0.03f_s$, phase margin
559 60° , $K_b = K_i/K_p$. *PID*: zeros cancelling the *LC* pair of $\Delta(s)$, crossover set for
560 60° of phase margin including the ESR zero, backward-difference derivative
561 on $-v_o$, $b = 1$, $c = 0$, $K_b = 1/\sqrt{T_i T_d}$. *PIDF*: the Type-III rule, i.e. the
562 PID zeros plus a filter pole on the ESR zero $N = N_C$, 60° of phase margin.
563 *LQR*: discrete Riccati solution on the ZOH model with the Bryson limits of
564 ξ_0 , measured i_L and $v_C = v_o - r_C i_C$, no integral action. *LQG*: the LQR gain
565 with a steady-state Kalman predictor on v_o only (process noise 0.01 in duty,
566 measurement noise 0.05 V). *LQI*: discrete Riccati solution of the integral-
567 augmented ZOH model with the same weights, measured state, $K_b = \omega_B$.
568 None of them is tuned by the proposed search; they represent what the
569 standard rules deliver on this plant.

570 4.6. Statistics and validation

571 Methods are compared as matched blocks over the 30 seeds with the
572 Friedman test and Holm-adjusted paired Wilcoxon signed-rank tests [17];
573 a run's score is J_s if feasible and 10^3 plus its violation otherwise, which is
574 consistent with Deb ordering. Families are compared pairwise on the same
575 (method, seed) cells. The retained controller of each family is the best feasible
576 record over its 180 runs; it is frozen and replayed on the 324 ms reference
577 mission of the Simulink model (24→1→46 V at 108 and 216 ms, +0.5 A load
578 pulse at 54–90 ms, initial state at the 24 V equilibrium), with the energy
579 balance of (8).

580 5. Results and discussion

581 5.1. Formula-first verification

582 Every analytical quantity of Sections 2–3 was recomputed twice: from the
583 closed forms and from independent numerical routes (state-space to transfer-
584 function conversion, a general CARE solver, a frequency sweep of (12) against

Table 7: Structural-equivalence audit (Proposition 2): largest $|v_o^{LQI} - v_o^{PIDF}|$ over the 324 ms mission.

Case	predicted	measured
nominal, $i_p = 0$	identical	1.6×10^{-13} V
$L - 20\%$, $r_L \times 2$, $R_{on} \times 2$	identical	2.8×10^{-14} V
0.5 A load pulse	$\neq$, Eq. (17)	0.411 V
$R + 20\%$	$\neq$, Eq. (17)	0.609 V
$C + 20\%$	$\neq$, $N \neq N_C$	2.823 V
sampled, ZOH filter, 1→46 V	close	14.54 V
sampled, FOH filter (19), 1→46 V	close	0.35 V

(13), and an SDP solver for (18)). The results of Section 3.5 agree with the values archived by the earlier version of this study, obtained with a different code base: the gain, the poles and θ_0 to all printed digits, and the LMI bounds α_{cert} , ζ_{cert} to within 2×10^{-7} relative, the bisection tolerance. The companion MATLAB script `cmo_formula_first.m` repeats the chain with `icare` and `freqresp`.

5.2. The equivalence, tested where it should hold and where it should fail

Table 7 and Fig. 5 test Proposition 2 on the 324 ms reference mission with continuous-time controllers on the nonlinear averaged plant (RK4, 0.05 μ s). With nominal output network and no load pulse the LQI and its 2-DOF PIDF produce identical trajectories to 1.6×10^{-13} V, although 1.1 % of the mission is spent in duty saturation, the averaged DCM guard $i_L \geq 0$ is active and the anti-windup path is exercised. Changing L by -20% and doubling r_L and R_{on} leaves the identity intact (2.8×10^{-14} V, with 34.1 % of saturation), exactly as the proof predicts, because these parameters never enter the output network. The three predicted failure modes appear with the predicted signs: a 0.5 A load pulse (0.411 V), a 20 % load mismatch (0.609 V) and a 20 % capacitance mismatch that detunes N from N_C (2.823 V).

Sampling is the practical case. With the zero-order-hold filter the sampled PIDF departs from the sampled full-state LQI by 14.54 V on the 1→46 V step; with the triangle-hold update (19) the departure falls to 0.35 V, while both sampled laws sit 2.54 V from the continuous trajectory because of the common one-sample delay. The realization is therefore exact in continuous time and remains a close realization under sampling *provided* the filter is discretized as the capacitor-voltage observer it is.

Table 8: CMO–LQI–PIDF family: 30 matched seeds per algorithm (Algorithms 1–6, each followed by refineBO). J_s statistics over feasible runs; “global counts runs already feasible after the 240 global calls.

Algorithm	feasible	global	best J_s	median	IQR	mean rank
PSO	18/30	18/30	0.8152	0.8283	0.0208	3.77
GA	13/30	13/30	0.8151	0.8330	0.0190	4.23
ABC	17/30	8/30	0.8152	0.8163	0.0162	3.77
pAEABC	25/30	18/30	0.8151	0.8161	0.0177	2.77
pIGWO	28/30	27/30	0.8153	0.8308	0.0205	3.10
pIGWO-DLH	23/30	18/30	0.8152	0.8291	0.0256	3.37

5.3. Seven algorithms on the LQI-weight space

Table 8 and Fig. 6(a) summarize the 180 runs of the CMO–LQI–PIDF family. 124 runs end feasible. All six branches converge to the same optimum: their best records lie between 0.8151 and 0.8153, a spread of 2×10^{-4} , and decode to practically the same controller. The LQI-weight space therefore has a single, well-defined optimum under the 33 inequalities, and the retained design is not an artefact of one operator or one seed. The retained record improves the normalized objective by 18.5 % with respect to the Bryson record ξ_0 , while ξ_0 itself violates the overshoot specification.

What separates the operators is reliability, not the optimum. The Friedman test over matched seeds rejects equality ($Q = 11.96$, $p = 0.035$); after Holm correction 1 pairwise difference remains (GA–pIGWO). pIGWO reaches a feasible design in 28 of 30 seeds and pAEABC in 25, against 13 for GA. The refinement stage matters most where the global stage is weakest: it raises the number of feasible runs from 8 to 17 for ABC and from 18 to 25 for pAEABC (Table 8), whereas it adds one run for pIGWO and none for PSO and GA. A practitioner who can afford only one run should therefore prefer the grey-wolf or adaptive-bee variant followed by refineBO; one who can afford a few seeds can use any of the six.

Retained controller. The retained weights are $\Delta_i = 10.842$ A, $\Delta_v = 3.301$ V, $\omega_B = 6505$ rad/s, giving $K = [0.090893, 0.165271, -945.806]$ and closed-loop poles -6493 and $-20149 \pm j17169$ rad/s. Its exact realization is

$$\theta^* = [K_p, K_i, K_d, N, b, c, K_b] = [0.17436, 945.81, 8.2629 \times 10^{-6}, 2 \times 10^5, 1, 0, 92951]. \quad (25)$$

Being an LQI law, it inherits the full-state LQR margins (minimum return difference 1, phase margin 70.25°), and the common- P certificate over the operating polytope gives $\alpha_{cert} = 6414.8 \text{ rad/s}$ and $\zeta_{cert} = 0.7496$, i.e. 1.54 times the guaranteed decay rate of ξ_0 at slightly better guaranteed damping. None of these certificates is available for a gain vector found by direct search.

5.4. Does the LQI parameterization help? Six families, one protocol

Table 9 and Fig. 6(b) compare six families run under the identical protocol, so that the only difference between two rows is the space in which the seven algorithms search.

Same structure, different coordinates. PIDF-direct searches the same 2-DOF PIDF ($b = 1$, $c = 0$, $N = N_C$) directly in $(\log K_p, \log K_i, \log K_d, \log K_b)$ over the gain ranges spanned by the LQI box. It reaches the same optimum (best $J_s = 0.8154$ against 0.8151) with a similar number of feasible runs (129 against 124 of 180), and the paired comparison on the same (algorithm, seed) cells is not significant ($p = 0.74$). The LQI coordinates therefore do not make the search easier on this problem. What they change is what every candidate is. Inverting the map (14) and applying Kalman’s return-difference condition shows that the best direct-search design is itself LQ-optimal, $K = [0.0906, 0.1650, -943.2]$ with $\min_\omega |1 + L_K(j\omega)| = 1$, i.e. the gain search converges onto the LQ manifold; yet only 75 of its 129 feasible designs satisfy the condition, so 42 % of the controllers an equally lucky gain search may return carry no LQ optimality, no guaranteed margins and no reason to be certifiable. The LQI parameterization confines the search to that manifold from the first evaluation, at no cost in optimality.

Richer and poorer structures. PID-direct (backward-difference derivative) reaches best $J_s = 0.8249$ in 125 feasible runs and is not significantly different from CMO-LQI-PIDF ($p = 0.41$). PIDF7-direct, with free filter pole and setpoint weights, finds more feasible runs (163) but a worse optimum (0.8513), with $N = 141939 \text{ rad/s}$, $b = 1.00$ and $c = 0.31$; on the switched mission this design holds the 1 V level with 0.0 mV of error and IAE 0.0348 V s, i.e. the extra freedom buys switched robustness at the price of the averaged objective. PI-direct never becomes feasible (minimum violation 1.04): a PI cannot meet the overshoot, load-deviation and current specifications of the six scenarios at once, and the paired comparison is decisive ($p = 2.7 \times 10^{-31}$).

Table 9: Controller families under the identical protocol (7 algorithms, 30 seeds, 300 calls, 33 inequalities). Paired comparison with CMO–LQI–PIDF on the same (algorithm, seed) cells: Deb-consistent score, two-sided Wilcoxon signed-rank test.

Family	variables	feasible runs	best J_s	median J_s (feasible)	CMO better / tie / worse	p	min. violation
CMO-LQI-PIDF	4 (LQI weights, K_b)	124/180	0.8151	0.8287	–	–	0.000
CMO-LQI-SF	4 (same, full state)	154/180	0.9054	0.9749	132 / 0 / 48	3.5×10^{-2}	0.000
PIDF-direct	4 (K_p, K_i, K_d, K_b)	129/180	0.8154	0.8273	99 / 0 / 81	7.4×10^{-1}	0.000
PIDF7-direct	7 (+ N, b, c)	163/180	0.8513	1.1018	121 / 0 / 59	7.7×10^{-1}	0.000
PID-direct	4	125/180	0.8249	0.8278	99 / 0 / 81	4.1×10^{-1}	0.000
PI-direct	3	0/180	–	–	180 / 0 / 0	2.7×10^{-31}	1.041

666 *The full-state twin.* CMO–LQI–SF uses the same four variables but applies
667 the gain as a measured-state LQI. It is feasible more often (154 runs) but
668 at a worse averaged optimum (0.9054) than its output-feedback realization
669 ($p = 0.035$ in favour of the PIDF on the averaged objective). On the switched
670 mission the ordering reverses (Section 5.5): the twin is the more robust
671 controller at the 1 V level, which is the switched counterpart of the output-
672 current term of (17).

673 *What the protocol, rather than the parameterization, buys.* Every family tuned
674 under the constrained multi-fidelity protocol ends close to full compliance on
675 the reference mission (Table 10: u between 0.74 and 1.02; the lowest value
676 belongs to a slow PI whose IAE, 0.0943 V s, is 2.6 times that of the others),
677 whereas the six fixed-rule designs lie between 6.28 and 30.17. The decisive
678 ingredients are the engineering-unit specification and the switched transition
679 screen; the choice of search coordinates decides the guarantees that come
680 with the result.

681 5.5. Switched validation on the reference mission

682 Table 10 and Fig. 7 report the frozen controllers on the 324 ms mission of
683 the reference model. The energy balance closes to better than 4.5×10^{-6} % of
684 the input energy for every row, so all efficiency and loss figures are resolved
685 by the integrator. As a fidelity check against the Simulink audit of the earlier
686 version of this study, the replica gives for the fixed PI 26.63 V of overshoot,
687 40.78 A and IAE 0.0480 V s (Simulink: 26.75 V, 40.76 A, 0.0464 V s) and for
688 the fixed LQI 9.41 V, 32.61 A and 0.0348 V s (Simulink: 9.56 V, 31.27 A,
689 0.0346 V s).

690 *Answer to the comparison question.* Against the six classical designs the
691 result is unambiguous on the specification, and nuanced on individual indices.

Table 10: Switched replay on the 324 ms reference mission (24→1→46 V, 0.5 A load pulse). Specification ratios: overshoot/1.5 V, load deviation/0.5 V, peak i_L /23 A, worst steady error/25 mV, steady duty peak-to-peak/0.20; a ratio ≤ 1 meets the specification. u is the largest ratio. Energy residual below 4.5×10^{-6} % for every row.

Controller	IAE (V s)	overshoot (V)	$i_{L,pk}$ (A)	$ e _{\max}$ (mV)	load dev. (V)	η (%)	E_{loss}/E_u	sat. frac.	u	specs met	sensors
CMO-LQI-PIDF (retained, v_o only)	0.0364	0.02	22.44	25.6	0.137	97.978	0.01801	0.089	1.02	4/5	v_o
CMO-LQI-SF (twin, measured i_L)	0.0347	0.02	22.87	0.0	0.214	97.978	0.01800	0.008	0.99	5/5	v_o, i_L, i_C
PIDF, direct gain search	0.0363	0.02	22.42	25.0	0.137	97.978	0.01801	0.089	1.00	5/5	v_o
PIDF7, direct search	0.0348	0.02	22.11	0.0	0.164	97.978	0.01800	0.011	0.96	5/5	v_o
PID, direct search	0.0368	0.02	21.83	23.7	0.138	97.980	0.01798	0.087	0.95	5/5	v_o
PI, direct search	0.0943	0.23	6.04	6.8	0.369	97.987	0.01786	0.009	0.74	5/5	v_o
Bryson LQI-PIDF ξ_0	0.0340	8.05	32.50	0.0	0.193	97.936	0.01845	0.012	5.37	3/5	v_o
PI (loop shaping)	0.0480	26.63	40.78	0.1	0.474	97.813	0.01972	0.012	17.76	3/5	v_o
PID (pole-zero)	0.0467	28.64	40.77	11.3	0.128	97.802	0.01984	0.093	19.09	3/5	v_o
PIDF (Type III)	0.0453	28.65	40.78	0.0	0.157	97.801	0.01986	0.013	19.10	3/5	v_o
LQR	0.1339	0.97	24.36	754.3	0.805	97.969	0.01811	0.001	30.17	2/5	v_o, i_L, i_C
LQG	0.0716	0.28	24.39	289.5	0.466	97.960	0.01818	0.000	11.58	3/5	v_o
LQI (discrete, full state)	0.0348	9.41	32.61	0.0	0.552	97.931	0.01849	0.013	6.28	2/5	v_o, i_L, i_C

- *PI, PID and PIDF* designed by standard frequency-domain rules hold every level but overshoot the 1→46 V step by 26.63–28.65 V and draw about 40.78 A, i.e. 1.8 times the 23 A device rating; their specification ratios are 17.76–19.10.
- *LQR and LQG*, having no integral action, leave offsets of 754.3 and 289.5 mV at the worst level.
- *LQI*, synthesized from the same Bryson limits and reading the full state, tracks well (IAE 0.0348 V s) but overshoots by 9.41 V and peaks at 32.61 A.
- The retained *CMO-LQI-PIDF*, reading v_o only, removes the overshoot, stays at 22.44 A, keeps the load deviation at 0.137 V and converts at 97.978 %, against 97.80–97.97 % for the six fixed-rule designs. Its IAE, 0.0364 V s, is within 5 % of the LQI and 7 % of ξ_0 , whose lower IAE is bought with 8–9 V of overshoot. It misses one of the five mission specifications by 0.6 mV: 25.6 mV of steady error at 1 V for a 25 mV limit ($u = 1.02$).
- Its full-state twin *CMO-LQI-SF*, with the same weights and a measured inductor current, meets all five specifications ($u = 0.99$) with IAE 0.0347 V s.

The procedure is therefore superior to the six classical design rules in the sense that matters to an industrial user: every controller it returns, in any of the tuned families of Table 9, removes the overshoot, stays within the

device rating and holds the levels, with specification ratios between 0.74 and 1.02 against 6.28–30.17 for PI, PID, PIDF, LQR, LQG and LQI. It does not dominate every index: the LQI and ξ_0 track slightly better in IAE, at the cost of 8–9 V of overshoot and 32–33 A of peak current. The retained CMO–LQI–PIDF and its direct-search counterpart are practically the same controller (IAE 0.0364 and 0.0363 V s), and their 1 V steady error sits on the 25 mV line (25.6 and 25.0 mV); this level is the binding one, which the switched screen resolves only to within a millivolt because it is a shorter transient than the mission.

The value of the current sensor. Among the LQI-structured controllers, the full-state twin meets every specification and the output-feedback realization misses one by 0.6 mV. By Proposition 2 the two differ only by the term $-K_{ix}(i_o - v_o/R)$ of (17), an instantaneous feedback on the unmodelled output current. At 1 V in discontinuous conduction the output-current estimate v_o/R is least accurate relative to the signal, and the PIDF must remove the resulting bias through its integrator in the presence of a switching limit cycle; the twin removes it directly. The gap between the two rows of Table 10 is thus the measured dividend of one current sensor, predicted in sign and location by the theory. The efficiencies of all tuned designs agree within 0.01 percentage point (97.978–97.987 %), so the conversion loss does not discriminate between them; it does separate them from the fixed-rule designs, whose overshoots and current peaks cost 0.01–0.18 percentage point.

5.6. Limitations

The evidence is simulated. The switched replica reproduces the controller-independent figures of the reference Simulink model and its fixed comparators, but the final numbers of Table 10 should be regenerated on the `.slx` with the provided S-function before being quoted as Simulink results. Losses are the modelled conduction and snubber losses of the declared boundary; dynamic switching loss, reverse recovery, magnetic and thermal effects are outside it, and the efficiency differences between rows (a few hundredths of a percentage point) are far below the accuracy of a laboratory power analyser. Component tolerances were not supplied and are not claimed: the common- P certificate covers the nominal operating polytope only. The mission and the specification are those of a fixed 48 V source; line regulation is outside the study. Finally, the specification set and the filter discretization were revised

during pilot campaigns that are reported in Section 4 and Appendix Appendix B; the final campaign was run once, with the protocol frozen, and its switched validation was not used for selection.

6. Conclusions

A formula-first co-design was carried out for a non-ideal asynchronous Buck converter supplied at 48 V and commanded over 1 V to 46 V. Its search space is the set of Bryson excursions of an LQI cost, so every evaluated point is a Riccati-optimal law with LQR margins and an operating-polytope certificate, and every point is implemented exactly as a 2-DOF PIDF that measures the output voltage only.

Three results carry the paper.

Theory. The realization is structural, not only nominal: the derivative filter of the PIDF, with its pole at the ESR zero, reconstructs the capacitor voltage along every trajectory of the converter, so the PIDF and the full-state LQI coincide under saturation, anti-windup and discontinuous conduction and for any inductor-branch or semiconductor parameters. They differ only by an explicit feedback on the unmodelled output current, by a mismatch of the output network, and by sampling. Under sampling the equivalence survives when the filter is discretized as the observer it is: a triangle-hold update keeps the sampled PIDF within 0.35 V of the sampled LQI on a 45 V step, where the usual zero-order-hold update departs by 14.54 V.

Optimization. Seven algorithms applied to the four LQI weights converge to one optimum; they differ in reliability, pIGWO and pAEABC reaching feasibility in 28 and 25 of 30 seeds against 13 for GA, and the Gaussian-process refinement matters most where the global stage is weakest. Searching the same 2-DOF PIDF directly in gain space reaches the same optimum, which turns out to be LQ-optimal; the LQI parameterization does not accelerate the search on this problem, but it confines it to the manifold on which the optimum lies and gives every candidate its margins and certificate, whereas 42% of the feasible direct-search designs are not LQ-optimal for any weight.

Validation. Switched admissibility is not inherited from an averaged model: an analytical ripple bound and a steady switched screen both admitted designs that limit-cycle on the reference mission, because the loop holds coexisting switching limit cycles selected by the preceding transient. A 34 ms switched transition screen inside the search removes them. On the reference mission, the retained CMO-LQI-PIDF removes the overshoot of

the classical designs and stays within the 23 A device rating; its only miss is a steady error of 25.6 mV at 1 V against a 25 mV limit, which its full-state twin meets. PI, PID, PIDF, LQR, LQG and LQI designed by their standard rules exceed at least two specifications by factors of 6 to 30.

The contribution is therefore a design procedure an industrial engineer can apply with standard tools: choose admissible excursions, solve one Riccati equation, realize the gain as a PIDF with setpoint weight one and a derivative pole on the ESR zero, discretize the filter as a capacitor-voltage observer, and let a small, constrained, multi-fidelity search tune four physically meaningful numbers. Hardware validation, component-tolerance robustness and line regulation are the next evidence levels.

Appendix A. Reproducibility record

Reference model. buck_fo_lqi_fixed_reference_v6_1_46V_R2022a.slx, SHA-256 948fbae2679b57ed726a92eeebcae398aaf0ca6b9b3a74216759ca80b390c7fe.

All plant, mission and logging parameters of Table 3 were decoded from its model workspace; the snubber is resistive ($R_s = 100 \text{ k}\Omega$, $C_s = \infty$) on the MOSFET only, the diode snubber is disabled, and the MOSFET body diode has $R_d = 0.01 \Omega$, $V_{fd} = 0$. The saved model initializes L at 1.2 A and C at 12 V; all replays of this paper start at the 24 V equilibrium (2.4 A, 24 V).

Commands. From the repository root:

```
python scripts/cmo_formula_first.py    # Sections 3-4, Prop. 2 tests
python scripts/cmo_campaign.py          # 6 families x 6 methods x 30 seeds
python scripts/cmo_analyze.py           # statistics, freeze, switched replay
python scripts/cmo_figures.py; python scripts/cmo_numbers.py
```

In MATLAB (R2022a or later, Simscape Electrical, Control System Toolbox):

```
cmo_formula_first
T = run_cmo_validation(slx, 'results/cmo/controllers.csv');
```

The second call replaces only the controller S-function (`cmo_sfun`, a line-by-line port of `cmo/ctrl.py`) and never saves the model.

Seeds and budgets. Seeds 2026091201, ..., 2026091230; population 12; 240 global and 60 refineBO calls per run; 30 runs per method and family. Each checkpoint stores the method, seed, best decision vector, objective, the 33 inequalities, the global-stage record and the best-so-far curve.

819 **Appendix B. Protocol changes with respect to the earlier version**

820 **Data availability**

821 All code (Python replica, evaluator, algorithms, analysis, figures), the
822 MATLAB S-function and replay script for the reference Simulink model,
823 the per-run checkpoints and the generated tables are archived with the
824 manuscript repository; a public DOI will be assigned at acceptance.

825 **Declaration of competing interest**

826 The authors declare that they have no known competing financial inter-
827 ests or personal relationships that could have appeared to influence the work
828 reported in this paper.

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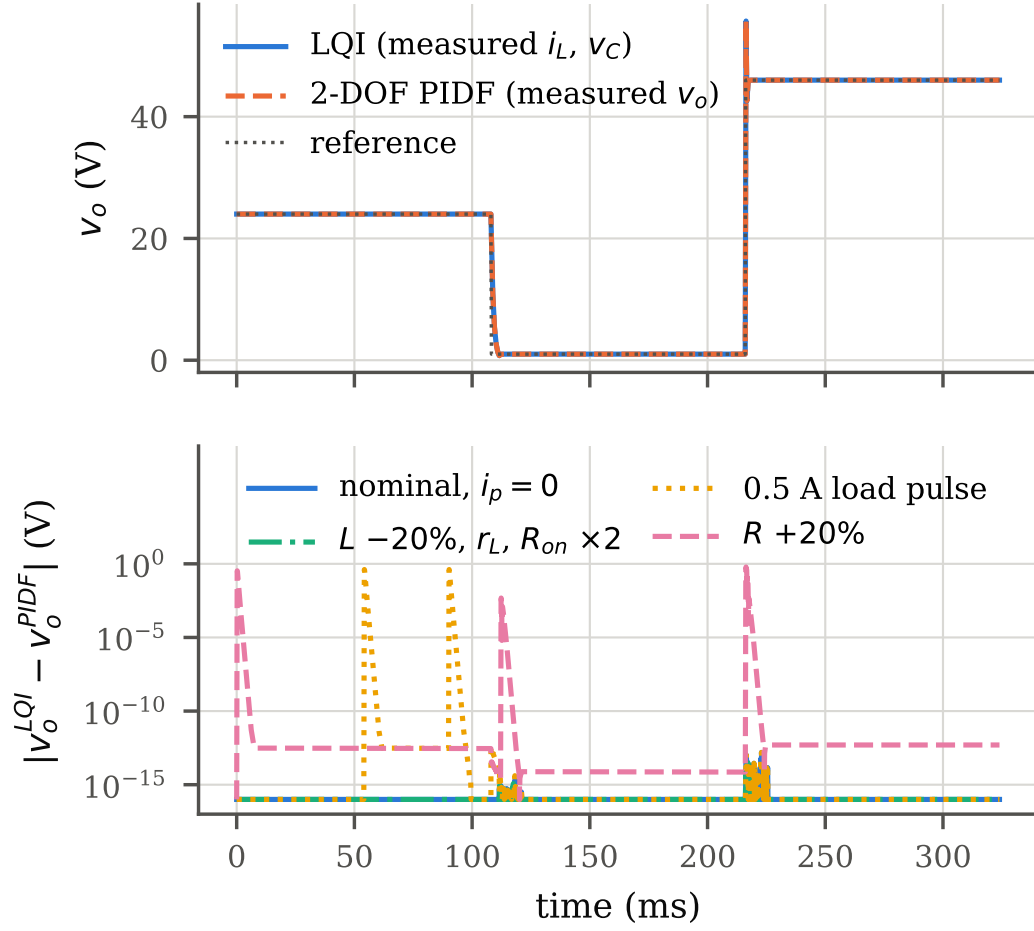


Figure 5: Top: LQI with measured (i_L, v_C) and its 2-DOF PIDF with measured v_o on the reference mission (continuous time, averaged nonlinear plant, saturation active). Bottom: absolute output difference; the nominal and inductor-branch-mismatch curves lie at the floating-point floor, the load-pulse and load-mismatch curves show the predicted departures.

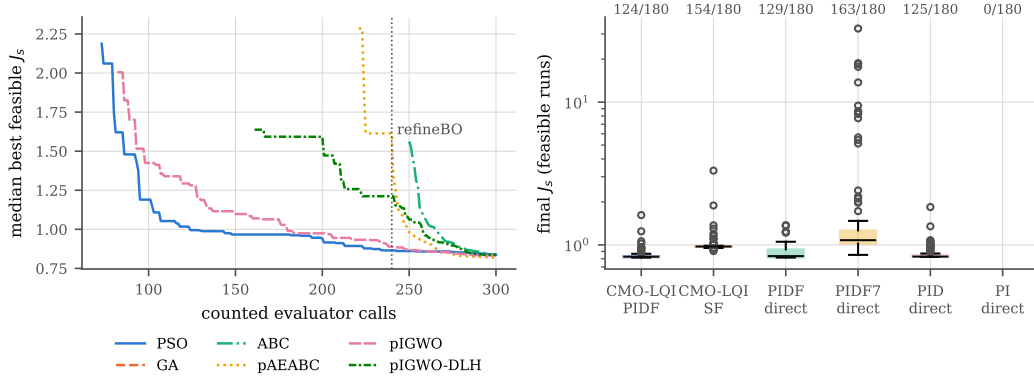


Figure 6: (a) Median best-feasible objective against counted calls for the CMO-LQI-PIDF family (dotted line: end of the global stage); a curve starts once at least half of the 30 seeds hold a feasible record, which GA never achieves (13/30). (b) Final objective of the feasible runs of every family under the identical protocol, with the number of feasible runs out of 180 above each box.

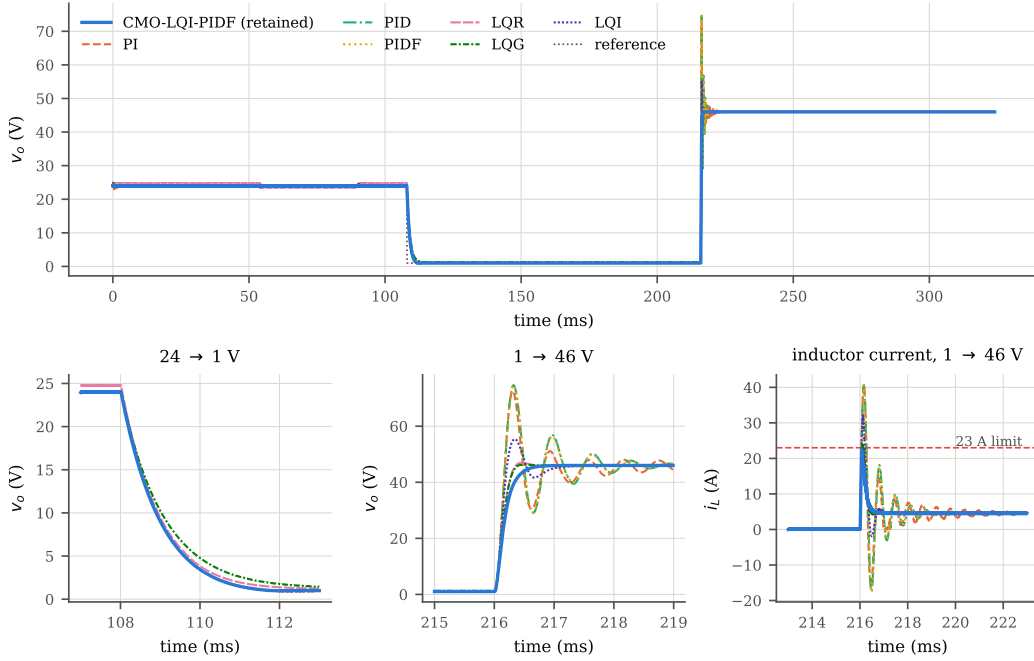


Figure 7: Switched replay on the reference mission: retained CMO-LQI-PIDF and the six fixed-rule comparators. Bottom: 24→1 V and 1→46 V transitions and the inductor current on the 1→46 V step against the 23 A device limit.

Table B.1: Declared changes and their reasons.

Change	Reason
Feasibility in engineering units instead of ratios to ξ_0	with ratio gates 24 of 36 constraints are active at ξ_0 , and a gentle record leaves no dominating candidate; absolute gates make all families comparable
Analytical ripple bound $\rightarrow$ switched transition screen (g_{31} – g_{33})	pilots 1 and 2: the bound and a steady screen both admitted limit-cycling designs (Section 4)
Device current limit 23 A (IRF540N at 100 °C)	the earlier retained design used 90 % of the 25 °C rating
refineBO budget stated once: 60 calls	the earlier text gave both 20 and 60
Structural equivalence (Prop. 2)	replaces the claim that saturation, DCM and PWM break the realization
Triangle-hold (FOH) discretization of the derivative filter, Eq. (19)	the ZOH filter makes the sampled PIDF deviate by up to 14.54 V from the sampled LQI on the 1 $\rightarrow$ 46 V step; FOH reduces it to 0.35 V (identified by the equivalence audit, before the final campaign)
Useful-energy band reported, not constrained	ill-posed on step-down transients of an asynchronous converter (Section 4); identified before the final campaign
Six added families and six fixed-rule comparators on one specification	separates parameterization from optimizer